A Hybrid Neural-Microfacet BRDF Model for Real-Time Rendering

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Abstract

Over the past decade, microfacet-based BRDF models have formed the foundation of real-time rendering pipelines. Despite their widespread use, they often fail to reproduce subtle appearance effects arising from complex light–surface interactions, which have led to the emergence of specialized physics-based models for specific optical phenomena (e.g., diffraction, iridescence, multilayers). Although more accurate, these models lose versatility and lack performance for real-time rendering. Recently introduced, neural models have demonstrated their ability to approximate BRDF reference data coming from measurements, simulations, or even complex shading networks. However, most current neural models require relatively large networks, making them costly for real-time rendering. In this paper, we introduce a hybrid model that combines a GGX-type microfacet model and a neural model to leverage the best features of both representations. The neural component corrects the appearance approximated by the microfacet component, allowing much smaller network than in existing neural models. We show that, at identical memory cost, our model approximates measurements better than state-of-the-art neural models for a low evaluation overhead compared to a microfacet-based model. Furthermore, our hybrid model remains easily editable by artists and benefits from an important sampling scheme, making it attractive for both offline and real-time rendering.

Keywords. Neural Networks, Appearance Modeling, Real-Time Rendering, Editable Appearance.

1 Motivation

The appearance of an object emerges from the interaction between the light and its surface. Modeling it accurately is at the core of computer graphics and remains an active research topic. The reflection of the light by the surface is described by a physical quantity known as the Bidirectional Reflectance Distribution Function (BRDF). In computer graphics, especially in a real-time context, an ideal BRDF representation should satisfy the following properties:

(i) *expressiveness*, to represent a large number of materials,

(ii) *robustness* when fitting measurements or simulated data,

(iii) *fast evaluation* with importance sampling support,

(iv) *low memory impact* for stored parameters,

(v) *editability* through meaningful parameters.

Over the past decade, microfacet-based BRDF models have formed the foundation of real-time rendering pipelines [1]. Despite their widespread use, they often fail to reproduce subtle appearance effects arising from complex light–surface interactions. This has motivated specialized

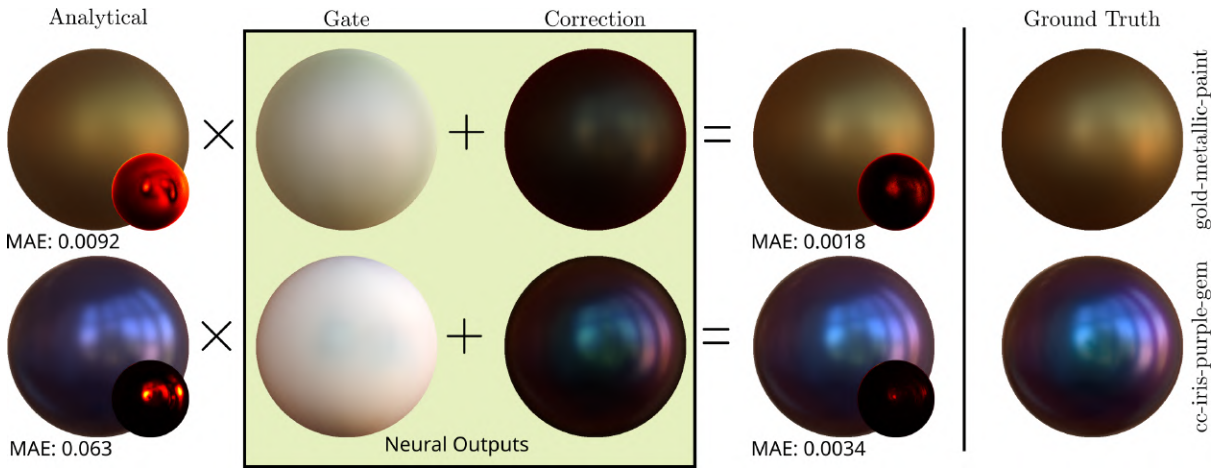

Figure 1: Our Neural BRDF model improves on a microfacet-based BRDF model using two modulation terms computed from a neural network: a multiplicative gate term and an additive correction term. Using a single shallow MLP (shared by many BRDFs) that can be evaluated in real-time we improve analytical models and outperform purely neural approaches for a similar compute budget. Our method also conserves the editability and importance sampling advantages from the underlying analytical model.

physics-based models for specific optical phenomena (e.g., diffraction, iridescence, multilayers). Although more accurate, these models lose versatility and generally lack performance for real-time rendering. Recently, neural models have shown strong ability to approximate BRDF reference data coming from measurements, simulations, or complex shading networks. However, most current neural models require relatively large networks, making them costly for real-time rendering.

In this paper, we introduce a hybrid model that combines an analytical microfacet-based model and a neural model to leverage the best features of each part. We then demonstrate that such a model has good characteristics regarding the above-mentioned requirements. The neural component acts as a corrective term, refining the output of the analytic BRDF to capture appearance effects that are otherwise difficult to reproduce. To keep memory tight **(iv)**, we enforce the learning of a common neural network for a whole collection of measured BRDFs, and only differentiate the correction term through a low-dimensional latent space as input of the network. Furthermore, we propose a joint training strategy that learns the parameters of both the analytic and neural components simultaneously on the whole BRDF collection **(ii)**, and demonstrate high-quality reconstruction **(i)** of many measured BRDFs from three databases (MERL [34], UTIA [15], RGL [13]).

We also enforce the reconstruction of the microfacet component to be close to the reference data during the training, implying that the network only learns residual effects rather than duplicating the base model. Restricting the network to predict residuals allows us to use a shallow multilayer perceptron (MLP) that remains small and computationally efficient **(iii)**, enabling high visual fidelity with minimal computational overhead, while still allowing the parameters of the analytic BRDF to be changed **(v)**. The reconstruction accuracy achieved by our model matches existing neural approaches while relying on a significantly smaller neural network, and our model can even outperform them when operating at comparable network sizes. As a result, it delivers clear improvements over standard analytical BRDF models while preserving real-time performance.

In the following, we review the relevant previous work on BRDF representations (Section 2), then introduce our hybrid model and the joint fitting procedure (Section 3). We then demonstrate its expressiveness capabilities by approximating a large number of real-world measurements

(Section 4). More precisely, we evaluate our model in term of BRDF reconstruction error and rendering error, comparing it to the neural state-of-the-art approach [54] (Sections 4.1 and 4.2). Finally, we demonstrate editing capabilities of our model (Section 4.3) and conclude by discussing the limitations and sketching future directions (Section 5).

2 Related Work

In the context of real-time rendering, building a representation that simultaneously satisfies the five criteria defined in the previous section remains particularly challenging. In the following, we review the main families of BRDF representations: tabulated approaches, analytic representations, non-parametric models, and neural ones.

Tabulated BRDFs are obtained thanks to dense measurements (e.g., [35]) or long light transport simulation (e.g., [19, 17, 52]) and permit very high quality reconstruction and fast evaluation. However, they are generally impractical in production rendering due to prohibitive memory requirements. A more practical approach regarding memory cost is to model the BRDF with a parametric function.

Single-Lobe BRDF Model most popular framework is the microfacet one, introduced to the Computer Graphics community by Cook and Torrance cookRefle82. This framework considers that the reflectance of a surface can be modeled as a collection of small Fresnel mirrors (i.e., microfacets) whose orientations are statistically defined by a distribution term (cf. [44]). Most of the time, the microfacet representation is supplemented by a Lambertian term. Although very efficient in terms of evaluation for real-time rendering [29] and easy to manipulate through its 7 or 8 parameters, the representation remains inherently limited in terms of expressiveness (cf. [36]). To overcome these limitations, different types of extensions have been proposed, such as correction for multiple-scattering between the microfacets (cf. [20, 27, 49] or to encapsulating wave phenomena (e.g.,[3, 16, 21]). Although these approaches significantly enrich the appearance of materials, they are usually effect-specific and increase both the number of parameters and the overall complexity of handling different materials. Another approach to augment expressiveness is to consider multiple-lobe parametric models.

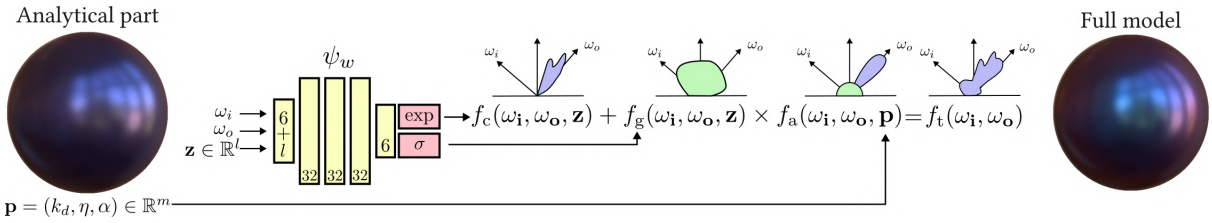

Figure 2: Overview of our method. We jointly optimize a latent code $\mathbf{z}$ and analytical parameters $\mathbf{p} = (k_d, \eta, \alpha)$ for each BRDF and a shared neural network. The output of the neural network (here, a 32x3 MLP) is split into two parts: the correction $f_c(\omega_\mathbf{i}, \omega_\mathbf{o}, \mathbf{z}) \in \mathbb{R}^3_+$ with an exponential activation and the gate $f_g(\omega_\mathbf{i}, \omega_\mathbf{o}, \mathbf{z}) \in [0, 1]^3$ with a sigmoid activation. These outputs are combined with the evaluation of the analytical model to produce our improved BRDF.
Overview of our method.

Multiple-lobe BRDF Models are either obtained empirically (e.g, [7, 28]) or by modeling the surface as a stack of parallel interfaces or media layers to obtain so-called layered BRDFs. The Weidlich and Wilkie weidArbit07 model, optimized later by Elek elek2010layered, was one of the first to provide an evaluation of the BRDF resulting from multiple layers. These models, which do not take fully into account scattering effects, are improved by the recent introduction of statistical models [10, 2, 51, 46, 39].

Despite these advances, the use of layered BRDF models remains limited since they require high memory demands (each layer requires its own set of parameters). In practice, to balance

quality, performance, and storage costs, real-time systems typically restrict the number and type of layers that can be represented. Moreover, although these models significantly increase the range of achievable appearances, they lack editability: the large number of interdependent parameters and the indirect relationship between model parameters and perceived appearance make intuitive control difficult. In particular, layered BRDFs require an explicit material decomposition that is hard to define and even harder to adjust a posteriori, as modifications to one layer propagate non-linearly through the entire model. Furthermore, as shown by Ngan Ngan2005, multiple-lobe models tend to be numerically unstable during the fitting process. For this reason, the aforementioned layered BRDF models are rarely tested against real-world BRDF measurements. Aside from parametric representations, non-parametric models have also been studied to represent BRDFs.

Non-parametric Models overcome physical constraints and seek and consider the BRDF as a signal defined on the hemisphere that can be projected into basis functions (e.g., Spherical harmonics [24, 41], wavelets [40, 8], Fourier basis [23, 53]) or reconstructed after a PCA on a database (cf. [35]). Other approximating approaches use radial basis functions [56], Spherical Gaussians [45, 50], Zernike polynomials [26], rational functions [37], Gaussian Mixtures [9], tensor products [6] or dictionaries [43]. These methods can represent all types of materials very precisely, as long as dense measurements or simulations are provided as input. Their main limitations are their memory cost, because it grows quadratically with the specularity [33] of the materials, and their lack of editing capabilities.

Neural methods have emerged as a powerful alternative to analytic BRDF models, predicting reflectance directly using neural networks. Such methods are effective in modeling even complex and layered materials [54, 14] and can also work as compression for measured materials [22, 42]. The use of autoencoders and hypernetworks [18, 5] allows for exploration of learned manifolds of BRDFs, enabling interpolation between data points or edition of the BRDF model through the latent variables [55]. Neural BRDF models do have some drawbacks compared to more physically based approaches: like most data-driven methods, they lack inductive bias for this task, thus requiring a large amount of data to train. Furthermore, when dealing with deep neural networks, the inference cost can quickly become prohibitive for real-time rendering. However, it has been shown that the use of shallow MLPs can be evaluated in real-time during the rendering stage [47, 54, 4]. Recent methods, such as [54, 12], explore the use of embeddings for real-time inference, reducing the size of the neural network but increasing the memory cost of the method.

Across these families of BRDF representations, none simultaneously satisfy the ideal BRDF representation for real-time applications, therefore motivating our hybrid approach.

3 Hybrid BRDF Model

In this section, we introduce a hybrid BRDF model that combines a standard analytical reflectance formulation with learned neural components, leveraging the strengths of both physically based and neural approaches. The idea is to let the analytical component model the dominant reflectance lobes and capture the primary perceptual characteristics of the material, while more complex and subtle effects are delegated to a neural network. We jointly optimize both the parameters of the analytical BRDF and the neural network weights in an end-to-end training process, allowing the model to automatically distribute representational capacity between the analytical and learned components.

3.1 General formulation

Our hybrid formulation models the target BRDF f_t as the combination of an analytical component f_a modulated by a gating function f_g and corrected by an additive residual term f_c:

$$f_\mathrm{t}(\omega_\mathbf{i}, \omega_\mathbf{o}) = f_\mathrm{c}(\omega_\mathbf{i}, \omega_\mathbf{o}, \mathbf{z}) + f_\mathrm{g}(\omega_\mathbf{i}, \omega_\mathbf{o}, \mathbf{z}) \times f_\mathrm{a}(\omega_\mathbf{i}, \omega_\mathbf{o}, \mathbf{p}), \tag{1}$$

where $\omega_{\mathbf{i}}$ and $\omega_{\mathbf{o}}$ are respectively the incident light and viewing directions, $z \in \mathbb{R}^l$ denotes a per-BRDF latent code and $p \in \mathbb{R}^m$ the per-BRDF parameters of the analytical model. The neural component of the model consists of both the gating function f_{g} and residual term f_{c}, which are jointly predicted by a single neural network ψ_w :

$$(f_{\mathrm{c}}, f_{\mathrm{g}}) = \psi_w(\omega_{\mathbf{i}}, \omega_{\mathbf{o}}; z)\,, \tag{2}$$

where w denotes the trainable parameters of the neural network. Figure 2 provides an overview of our approach and illustrates the structure of the proposed hybrid BRDF representation. We additionally enforce f_{c} to be positive, and constrain $f_{\mathrm{g}} \in [0, 1]$ using an exponential activation for the residual term and a sigmoid activation for the gating term. These constraints ensure that the final BRDF remains positive even for combinations of latent codes and analytical parameters not seen during training. This structural property enables intuitive material editing as further detailed in Section 4.3.

In this hybrid formulation, each term plays a distinct role. The analytical term f_{a} models the primary reflectance behavior and ensures that the model remains grounded to physically-based parameters. The gating function f_{g} provides a direction-dependent modulation that adjusts the influence of the analytical component. Finally, the additive correction term f_{c} enables the model to reproduce complex reflectance effects that the analytic model cannot express.

3.2 Model Architecture

In practice, our hybrid model formulation requires defining both a neural network architecture ψ and an analytical model f_{a}. To meet real-time rendering requirements, we restrict ourselves to a single-lobe analytical model together with a lightweight MLP for the network.

Analytical model Inspired by widely used models in real-time scenarios, we use an analytical component f_{a} composed of a Lambertian lobe and a GGX [44] microfacet-based specular lobe:

$$f_{\mathrm{a}}(\omega_{\mathbf{i}}, \omega_{\mathbf{o}}; p) = \frac{k_d}{\pi} + \frac{F(\omega_{\mathbf{i}}, \omega_{\mathbf{o}}, \eta) G(\omega_{\mathbf{i}}, \omega_{\mathbf{o}}, \alpha) D(\omega_{\mathbf{i}}, \omega_{\mathbf{o}}, \alpha)}{4(\omega_{\mathbf{i}} \cdot n)(\omega_{\mathbf{o}} \cdot n)}, \tag{3}$$

where the set of parameters $p = \{k_d, \eta, \alpha\}$ is to be fit per material, with $k_d \in [0, 1]^3$ being the diffuse albedo of the Lambertian term, $\eta \in [1, 10]^3$ corresponds to a colored index of refraction and model how the light is reflected with the Fresnel term F, and G and D are respectively the geometric attenuation and normal distribution terms of the microfacet model that depend on a roughness parameter $\alpha \in [0, 1]^2$ for anisotropic materials. We choose this analytical formulation for its simplicity and its widespread use in real-time rendering. In practice, we observe that across the 312 BRDFs fitted in our experiments, this single-lobe model combined with our neural correction is sufficient to reproduce the vast majority of target appearances accurately.

Network architecture As illustrated in Figure 2, we model both f_{c} and f_{g} with one shallow MLP ψ_w. In practice, we use an MLP with a few hidden layers (1-3) and with limited hidden dimensions (16-32) because such architectures can be evaluated in a fragment shader during the rendering stage for real-time applications [47], and optimized leveraging cooperative vectors [4]. The inputs of ψ_w are the incoming and outgoing directions $\omega_{\mathbf{i}}, \omega_{\mathbf{o}}$ stored in Cartesian coordinates and the latent code z for the selected BRDF. The last linear layer is split into two parts to which we apply an exponential activation for the residual term f_{c} and a sigmoid activation for the gating term f_{g}.

3.3 Fitting Method

We now describe how the parameters of our hybrid BRDF model are fitted to measured BRDF data. The parameters describing our models consist of the latent code $z \in \mathbb{R}^l$, the analytical

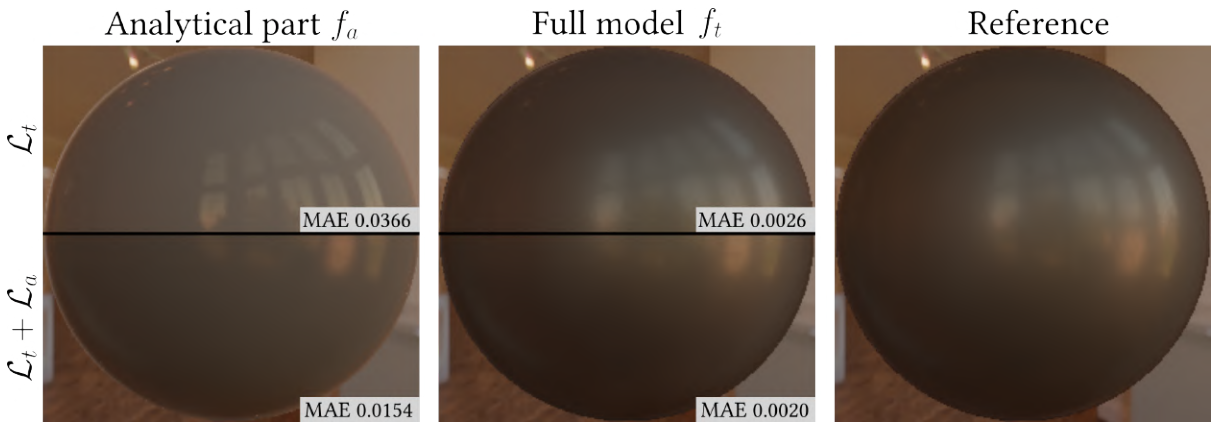

Figure 3: Impact of the analytical loss $\mathcal{L}_a$ term. **Top-part**: with full loss (4), f_a remains close to the target reflectance. **Bottom-part**: without $\mathcal{L}_a$, f_a is allowed to deviate from the target. This impacts the overall quality and weakens the editability, as the analytic parameters p no longer match the final appearance. Tested Material: `alumn-bronze` from MERL.

parameters p and the neural network weights $w \in \mathbb{R}^m$. Although it would be possible to fit all these parameters independently for each target BRDF, this would not be practical in a real-time rendering context, as it would require storing a dedicated set of network weights w for every material. To avoid this prohibitive cost, our fitting strategy relies on sharing the network weights w across a collection of BRDFs $\{f_\mathrm{t}^i\}_{i=1}^N$.

In practice, we rely on a mini-batch stochastic gradient descent to jointly optimize the common neural network weights w together with the per-BRDF analytical parameters $\mathbf{p} = \{p_i\}$ and the per-BRDF latent codes $\mathbf{z} = \{z_i\}$. At each training iteration, we randomly sample a batch $\mathcal{B} = \{(\omega_i, \omega_o)\}$ of 1024 incident and outgoing direction pairs drawn from a cosine-weighted distribution over the hemisphere. Each BRDF f^i in the dataset is evaluated at these directions, and we minimize the following objective:

$$\sum_{i=1}^{N} \sum_{\omega \in \mathcal{B}} \mathcal{L}_a\Big(f^i(\omega), f_a(\omega; p_i)\Big) + \mathcal{L}_t\Big(f^i(\omega), f_t(\omega; p_i, z_i, w)\Big)\,, \tag{4}$$

where f^i, f_a, and f_t denote the reference, analytical, and hybrid BRDFs, respectively. The analytical loss term $\mathcal{L}_a$ encourages the analytical component to approximate the target BRDF as accurately as possible on its own, whereas the hybrid loss term $\mathcal{L}_t$ supervises the full model. Including both terms in the objective is essential to obtain the desired behavior: the analytical model captures the primary perceptual characteristics of the material, such as the dominant diffuse and specular lobes, whereas the neural network focuses on modeling complex residual effects that cannot be well approximated by the analytical formulation alone. Figure 3 demonstrates this behavior on an example material with the result obtained with the combined loss (4) compared to the result obtained with solely the hybrid loss term $\mathcal{L}_t$. Both loss terms use the same error metric:

$$\mathcal{L}_{a,t}(f_1, f_2) = \left\|\log(1 + \cos(\theta_i)\, f_1) - \log(1 + \cos(\theta_i)\, f_2)\right\|^2\,, \tag{5}$$

where θ_i denotes the incident elevation angle. The logarithmic compression reduces the dynamic rangewhile the cosine elevation downweights high-energy grazing-angle samples, which are known to dominate the loss and destabilize training [32].

In our implementation, the gradients are then computed using auto-differentiation from PyTorch [38], and optimization is done with the AdamW optimizer [25, 31]. In practice, we optimize the parameters w of the neural network ψ and the per BRDF parameters z, p starting

with a learning rate of 0.005 and a cosine decay for 200k steps. We find that clipping gradient norms over 0.01 helps with stability. A full training for 100 BRDFs (such as the MERL dataset) takes about 10 minutes on an RTX 5080.

3.4 Rendering Implementation

Integrating our hybrid model into a rendering system is straightforward. At runtime, the analytical component is handled exactly as in a conventional renderer, with the only addition being the inclusion of the latent vector in the material definition. The neural network is exported as a shader function that is evaluated after the analytical BRDF computation and outputs the gating and corrective terms, which are then used to update the analytical BRDF value according to Eq. (1). As a result, the memory overhead of the hybrid model is limited to the storage of the latent vector, while the size of the neural network (14 kB for a 32×3 MLP shared between all BRDFs of the dataset) determines the performance overhead. Both can be adjusted to meet the performance and quality requirements of the target rendering application.

Unlike recent neural approaches that require evaluating an extra network to predict either the sampling PDF [48] or the parameters of a proxy with a closed-form PDF [54], our hybrid model can be sampled by taking advantage of its analytical component f_{a}. Indeed, we apply MIS between cosine sampling and GGX lobe sampling without the need for any additional neural network evaluation (cf. Figures 8 and 9).

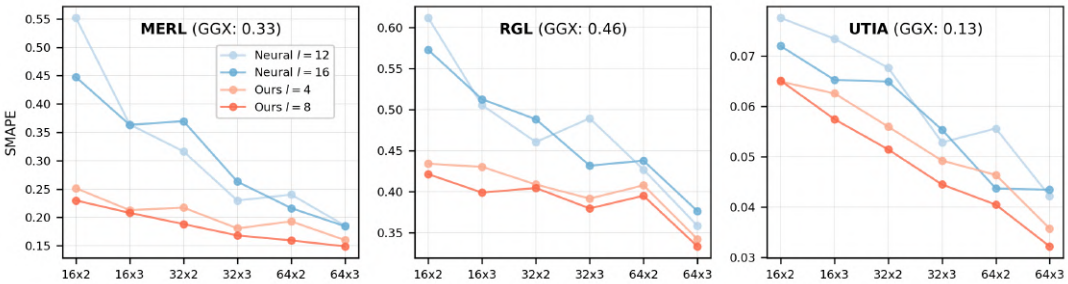

Figure 4: Average SMAPE on BRDF reconstruction for the 3 datasets for multiple MLP size. Since the analytical part of our hybrid model requires 7/8 parameters, we compare it to a fully neural model with 8 additional latent dimensions to provide a comparison with the same number of parameters (per BRDF) for both methods. The metric next to the dataset name is the SMAPE metric of the analytical part of the hybrid model on the particular dataset. For the MERL and RGL datasets, our approach, for small-sized MLPs, outperforms the Neural model or the GGX microfacet-based model. The improvement is not as significant in the case of the UTIA dataset because its materials are very similar, and the anisotropic GGX model alone already provides a good approximation.

4 Results

This section provides numerical results that demonstrate our model's ability to fit complex measured BRDFs (section 4.1). We evaluate the the rendering performance in Section 4.2) and provide experiments showing the robustness of the model to create coherent and plausible new BRDF outside the training set by editing the underlying parameters of the analytical model (section 4.3).

We evaluate our method on the following datasets: 100 Isotropic BRDFs from MERL matuDatad03, 51 Isotropic and 11 anisotropic BRDFs from the RGL Dupuy2018Adaptive, and 150 anisotropic BRDFs from the UTIA BRDF dataset filip14template. These three datasets provide a diverse collection of measured real-world materials covering a broad range of appearances, such as fabrics, metals, plastics, paints, etc.

In the experiments presented in this section, we trained our model on each dataset, using a single network shared by all BRDFs within that dataset. This highlights the ability of our hybrid model to represent a large variety of BRDF behaviors using a common network.

4.1 Quality analysis

To assess the quality of our model, we use two complementary metrics: one measuring the BRDF reconstruction error in function space and the other measuring the rendering error on a reference scene. We then compare these values to the one obtained with our implementation of the fully neural BRDF architecture from [54], and we refer to this method as *neural* in the following. This architecture consists of a shallow MLP together with a frame transform module that outputs 3 shading frames to transform the input directions. To be comparable, we replace the encoder part with a trainable latent code with a dimension that matches the dimension of the concatenation of our latent code z and the parameters of the analytical model p. We therefore consider a latent space of dimension 12 (resp. 16) to compare with our approach, with a latent code of dimensions 4 (resp. 8) since we also require 7 (8 for anisotropic) parameters for the analytical model.

BRDF reconstruction quality The BRDF-space reconstruction error is computed with the Symmetric Mean Absolute Percentage Error (SMAPE) by sampling both the target and the estimated BRDF with 10^6 directions $\mathcal{B} = \{(\omega_\mathbf{i}, \omega_\mathbf{o})\}$.

$$\mathrm{SMAPE}(f, \hat{f}) = \frac{2}{10^6} \sum_{\omega \in \mathcal{B}} \frac{|f(\omega)\cos(\theta_i) - \hat{f}(\omega)\cos(\theta_i)|}{|f(\omega)\cos(\theta_i)| + |\hat{f}(\omega)\cos(\theta_i)|} \,. \tag{6}$$

Figure 4 presents the average SMAPE value obtained across all BRDFs of each tested dataset for various MLP sizes both for our hybrid approach and the neural approach [54]. This clearly shows that our hybrid approach provides better BRDF reconstruction, especially for smaller models. In practice, this means that our model can match the quality of fully neural models at much lower computational cost, resulting in a higher frame-rate in real-time applications or shorter rendering times in path-tracing. As the network size increases, the gap between the two approaches decreases. This is expected since larger networks imply better expressiveness and can approximate the BRDF sufficiently well on their own. The relative contribution of the analytical part is therefore less important. Note that this also explains the smaller difference observed on the UTIA dataset, where even very compact networks already achieve low SMAPE values (roughly 10× smaller than on the other dataset).

Rendering quality Based on the Mean Absolute Error (MAE), we evaluate our model by comparing its renderings with those obtained with only its analytical component, and the neural model from [54]. These renderings are computed with 1024 samples per pixel (spp) and compared against a reference image computed with the measured BRDF. Figure 5 shows results for one material of each of the considered datasets in a simple test scene consisting of a sphere lit by the `Uffizi` environment map. The number of parameters (12) and the MLP size (32x3) are the same for the neural model and ours. As shown in the Figure 5, our model produces a color-averaged MAE lower than the ones from the Neural or GGX models. Table 1, which gathers the results obtained for the 300 materials tested as well as different MLP sizes and parameters, confirms the performance of our model in the vast majority of cases, especially for small MLPs. In summary, with small MLPs (16×3 and 32×3), our model consistently outperforms the neural model with the same network sizes, for a comparable rendering cost and identical memory usage. In practice, we found that the 16×3 and 32×2 networks offer a good compromise between expressiveness and evaluation time.

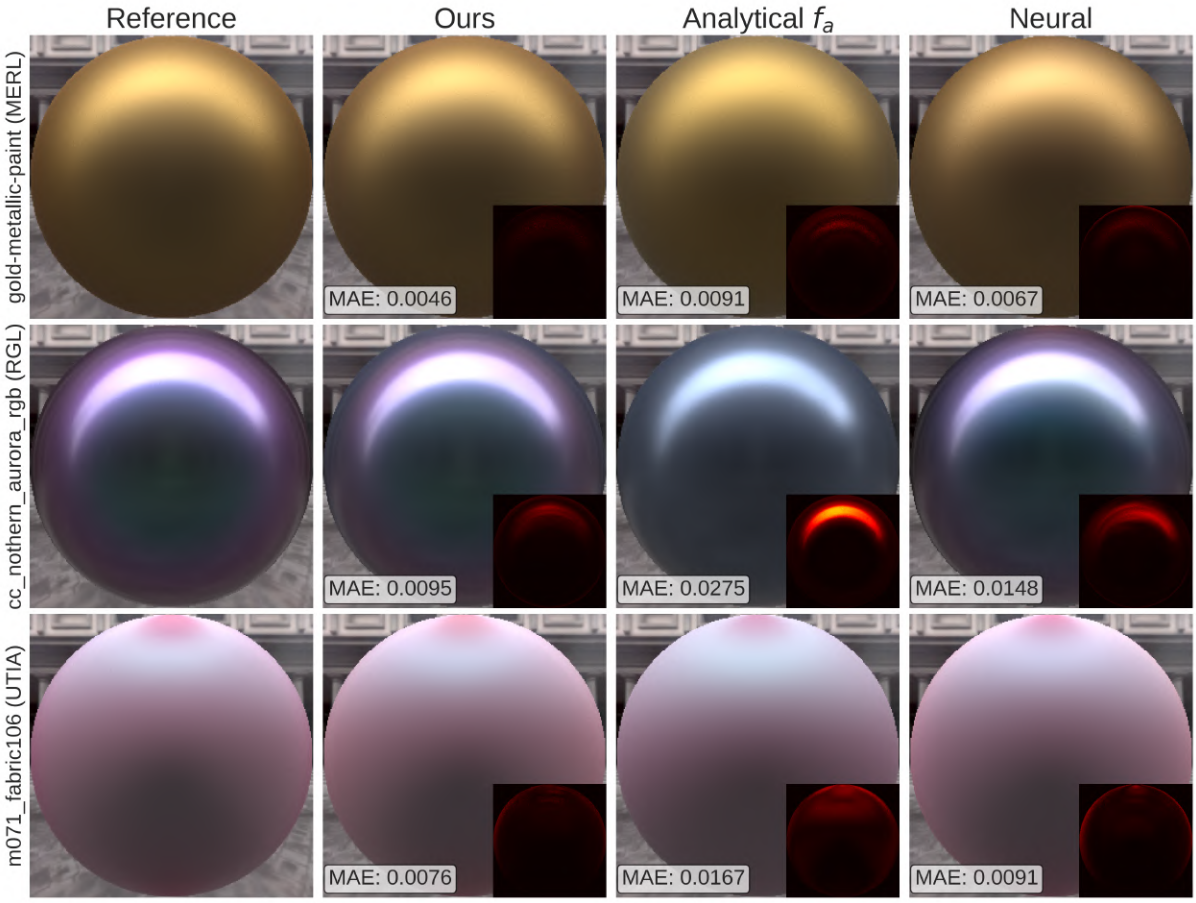

Figure 5: Per-material rendering comparisons between our model, its underlying analytical component, and the neural model. The insets show the per-pixel absolute error as well as the color-averaged MAE for the whole image. The number of parameters (12) and the MLP size (32x3) are the same for the neural model and ours.

Table 1: Mean Absolute Error (MAE) in terms of rendering. For the three datasets, we compare rendered images (same configuration as the ones from Fig. 5) between a GGX microfacet-based model, our hybrid model, and the Neural one for the same number of parameters and for different sizes of MLP. For a given dataset, the MAE is averaged over all renderings. Except for the RGL dataset and an MLP size of 64x3 or 16x3, our hybrid model exhibits better results.

MLP	**MERL** (GGX MAE: 0.0099)				**RGL** (GGX MAE: 0.0333)				**UTIA** (GGX MAE: 0.0201)			
	12 params		16 params		12 params		16 params		12 params		16 params	
	Ours $_{l=4}$	Neural $_{l=12}$	Ours $_{l=8}$	Neural $_{l=16}$	Ours $_{l=4}$	Neural $_{l=12}$	Ours $_{l=8}$	Neural $_{l=16}$	Ours $_{l=4}$	Neural $_{l=12}$	Ours $_{l=8}$	Neural $_{l=16}$
16×3	0.0059	0.0086	0.0056	0.0084	0.0267	0.0256	0.0245	0.0288	0.0109	0.0123	0.0106	0.0119
32×3	0.0051	0.0062	0.0052	0.0067	0.0203	0.0238	0.0203	0.0208	0.0101	0.0106	0.0098	0.0113
64×3	0.0046	0.0049	0.0044	0.0048	0.0149	0.0143	0.0160	0.0153	0.0094	0.0098	0.0093	0.0100

4.2 Rendering Performance Evaluation

We measure the impact of evaluating a MLP on the BRDF evaluation in BRDFExplorer [11]. Table 2 reports the frame rendering time when using the analytical BRDF alone, our hybrid BRDF, and the neural BRDF baseline for various MLP sizes. The evaluation time of our approach is comparable to that of the Neural model from [54] at equal network size. However, as shown in the previous section, our model achieves significantly better reconstruction quality for the same MLP configurations.

To evaluate the effectiveness of our importance sampling method (i.e., MIS between cosine

Table 2: Evaluation time (ms) of our hybrid model and the neural baseline for different MLP sizes, implemented in a GLSL shader in *BRDFExplorer* under environment map lighting, averaged over 100 spp, without importance sampling. Timings were recorded using NVIDIA Nsight on a RTX 5080.

MLP Size	16×2	16×3	32×2	32×3	64×2	64×3	GGX
Hybrid (ms)	0.10	0.13	0.22	0.37	0.78	1.40	0.03
Neural (ms)	0.16	0.17	0.30	0.45	0.98	1.59	

and GGX lobe sampling), we compare it to a tabulated reference method (similar to [30]). The tabulated representation is sampled at an angular resolution of 1 degree for a memory cost of 5.62 MB per material. Since this method samples the product of the BRDF and cosine terms of the rendering equation, it provides, at high angular resolution, the achievable limit in terms of sampling efficiency. To isolate the comparison of the importance sampling strategies, all BRDF materials are fitted with our model. Then we compare the tabulated method, a cosine-based one, and our method for the same number of samples per pixel (spp). As shown in Figure 9, the MAE between a converged 16K spp rendered image and our important sampling method is lower than with cosine sampling. Moreover, it tends to approach that of the tabulated method, while requiring significantly less memory. This is further shown with the numerical results of Figure 8 that represent the color-averaged variance of the absolute error in image space, with respect to the number of samples per pixel. These analyses show that the analytical component carries most of the BRDF energy, and it is therefore sound to use it as the primary source for importance sampling.

We integrated the hybrid BRDF model into a GPU path-tracer. We focus on models with a latent vector of size $l = 4$ since this configuration introduces minimal memory overhead. Using a compact 32×3 MLP, our path tracer renders a 1920×1080 frame with one sample per pixel and up to seven ray–surface interactions in approximately 47 ms, with runtime scaling as expected with the number of bounces. However, the analytical BRDF provides a natural fallback that can be exploited to reduce computational cost with limited impact on image quality. We evaluate a strategy where neural inference is disabled after a fixed number of bounces. Figure 11 shows results accumulated over 8000 spp to ensure convergence, while progressively disabling the neural correction after increasing numbers of interactions. The results indicate that after a small number of bounces, the neural component has a negligible visual impact. Leveraging this observation reduces the time to render one frame to 17 ms (on a RTX 4070Ti) by skipping neural evaluation after two bounces. This shows that the hybrid formulation is well-suited for real-time rendering, where neural inference can be selectively enabled only when it meaningfully affects appearance. Note that our current implementation is not fully optimized and does not exploit recent hardware accelerators such as cooperative vectors.

4.3 Editability

As our model acts as a correction operator over an analytical BRDF model, we can edit "non-destructively" the parameters of the underlying analytical model to explore possible appearances. Figure 6 shows an example of editing the parameters of the model we have defined in section 3.2 (i.e., diffuse color, refractive index, and roughness). When editing these parameters, we observed that the resulting BRDF remains physically coherent. This behavior is largely because the neural network provides a refinement of the analytical model, while the dominant reflectance information is encoded in the analytical model, thereby preserving the overall structure and physical validity of the reflectance behavior. It is the same reasoning that explains why two BRDFs can be interpolated in a consistent manner (Figure 10). Additional figures showing the edition and interpolation of BRDFs are available in the supplementary material. We also show,

as a toy example in the supplementary material, that it is possible to edit an isotropic material to make it anisotropic by replacing the normal distribution function of the underlying analytical model with an anisotropic one.

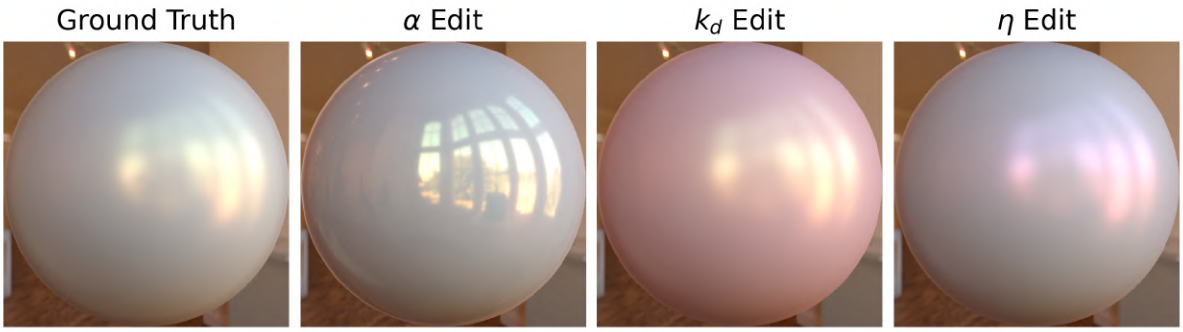

Figure 6: Editing the `satin-gold` BRDF from the RGL dataset by modifying the fitted parameters (roughness α, albedo k_d, IOR η) of the analytical component of our model. The Environment map is Wells6.

5 Discussion and Future Work

Model limitations The analytical part of the model is intrinsically interpretable from the microfacet theory and provides artistic control and editability (Section 4.3). However, the latent space that drives the corrective layer is not itself interpretable, and editing the latent vector after the optimization process is not guaranteed to preserve BRDF-like behavior. Moreover, our optimization process does not enforce any physical constraints such as energy conservation or Helmholtz reciprocity. However, we empirically observe that our learned BRDFs remain close to reciprocal in most cases (i.e., absolute error $< 1e-4$), and that their integrated energy tends to be similar or slightly lower than the reference. These limitations suggest interesting future directions: for instance, by adding physically based priors into the model or introducing regularization terms to better structure the latent space.

Multiple BRDF approximations In this work, we have shown that a single neural network can correct hundreds of BRDFs without having to distinguish between their types (fabrics, metal, dielectric, etc.). Introducing such distinctions would likely enable better reconstruction quality, but quantifying the improvements under different scenarios will need to be explored carefully. Beyond generalization across material types, this hybrid formulation also offers practical advantages for appearance digitization. Indeed, neural methods are known to perform well when fitting noisy data due to the stochastic nature of the training process, while analytical models perform well on very sparse data due to their built-in inductive bias. In our model, the analytical component acts as a regularizer, constraining the learned corrections to keep a physically meaningful reflectance behavior. This would be particularly beneficial for SVBRDF acquisition setups using real data from multi-view and multi-light setups.

Iridescent materials and Multi Lobe BRDFs The current implementation of our model may struggle to accurately represent complex effects such as iridescence (Figure 12). More generally, it tends to underperform on BRDFs that fall far outside the single-lobe framework with smooth angular variation. This is due to the strong single-lobe bias imposed by the GGX model used in the analytical component (Section 3.2). Capturing more complex effects would require increasing the expressive power of the model, either by enriching the analytical model or by relying on a more expressive neural component. Richer analytic models, such as multi-lobe

formulations, are significantly harder to optimize and introduce additional hyperparameters, while a larger neural component would allow to recover more complex appearances, at a substantial performance cost. In other words, finding the right balance between analytical priors and neural expressiveness remains an open challenge.

6 Conclusion

We have introduced a hybrid model that combines an analytical microfacet-based GGX model with a neural network. The latter precisely and automatically fine-tunes the analytical model to reproduce appearances that are not representable by the analytical model alone. We have shown that a shallow neural network, coupled with an analytical model, can reproduce complex appearances from various measured databases, while incurring only a slight increase in computational cost. Thanks to the analytical part of the model, our hybrid model remains easy to edit and benefits from the importance sampling scheme of the analytical component. Overall, this results in a new practical reflectance model that jointly offers high expressiveness, stable behavior, fast evaluation, low memory footprint, and intuitive editability, making it well-suited for production use.

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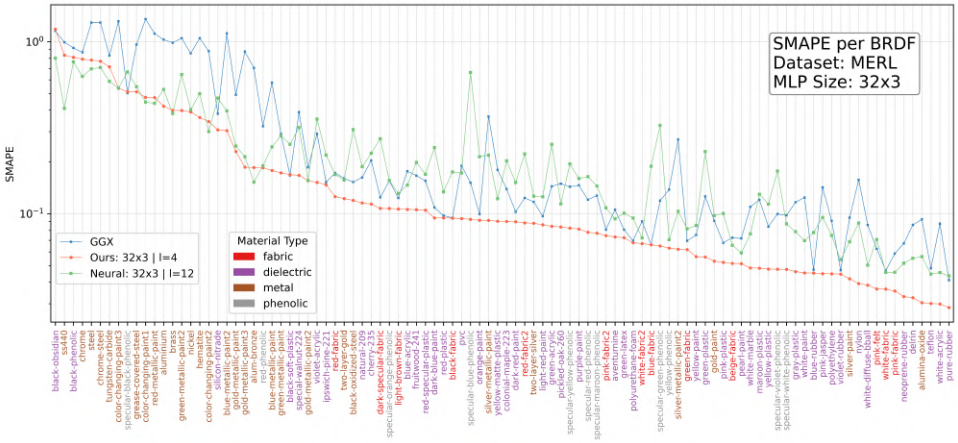

Figure 7: Per BRDF SMAPE on the MERL database, which consists of 100 measured materials. Our model with the 32x3 MLP and a 4D latent space is compared to the Neural model [54] with a 32x3 MLP and a 12D latent space, as well as a microfacet-based GGX model. Of the 100 materials, our hybrid model outperforms the Neural model for 88 materials of them. The supplemental material provides the same analysis for the RGB and UTIA databases.

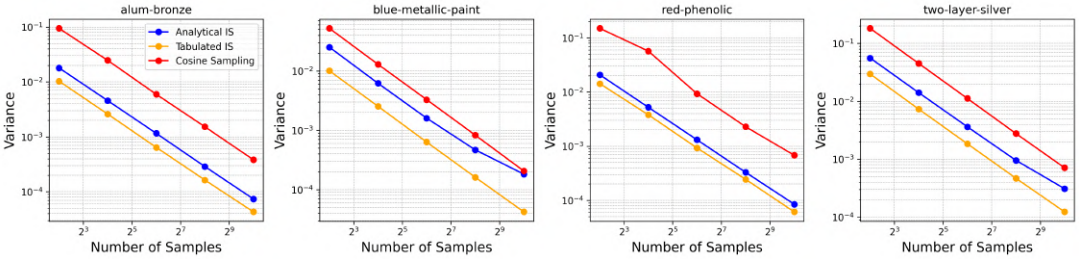

Figure 8: Color-averaged variance of the absolute difference between an image rendered with importance sampling and a reference image computed with 16k samples. Analytical Importance Sampling corresponds to applying either cosine sampling or GGX sampling using a MIS procedure for our model. As shown above, using our IS method approximates the true PDF quite accurately compared to the reference solution provided by tabulated Importance Sampling.

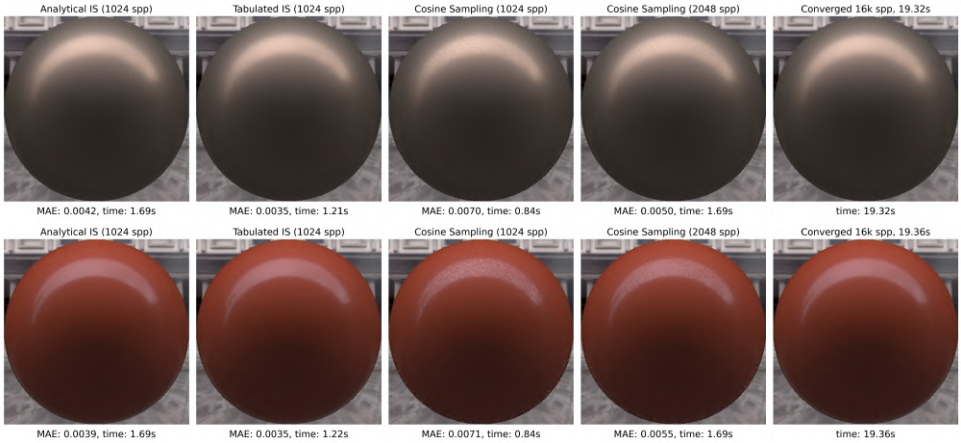

Figure 9: Importance Sampling (IS) comparisons in image space for the `red-phenolic` material (MERL). **From left to right:** IS based on the GGX distribution and a cosine lobe distribution, tabulated IS (similarly to [30]), with a memory footprint of 5.62 MB, the cosine lobe IS, and a converged result using tabulated IS. For all images, the BRDF values are taken from our fitted hybrid model. As shown by the two images on the left, using an importance sampling based on the microfacet model is not as optimal as the tabulated form but provides clear visual improvements over cosine lobe importance sampling.

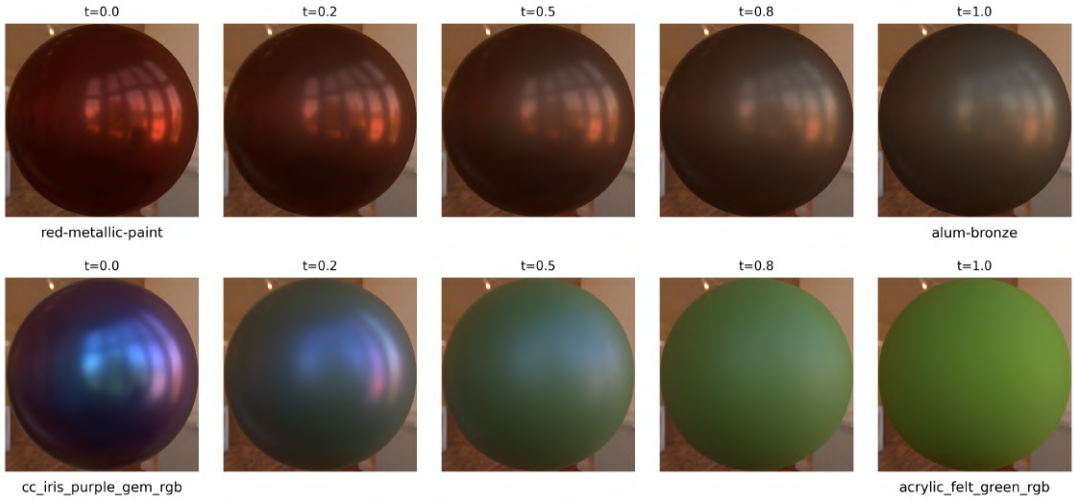

Figure 10: BRDF interpolation example. **From left to right**, we linearly interpolate (parameter t above) between two materials from (Top) the MERL dataset, or (Bottom) RGL dataset, the latent vector, and the analytical parameters of our hybrid model, showing very plausible results.

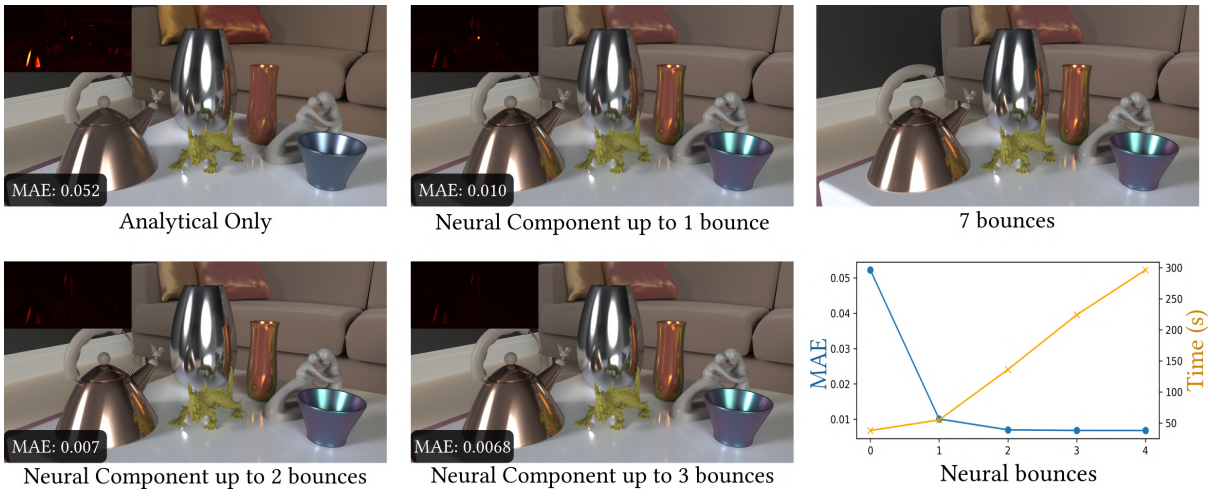

Figure 11: Path-Tracing performance evaluation. All materials of the scene's objects are rendered using our hybrid microfacet-neural model, which is fitted from three datasets. We show the effect in terms of MAE and rendering time, of falling back on the analytical component model after a certain number of bounces. The different materials used for this scene are listed in the supplementary material.

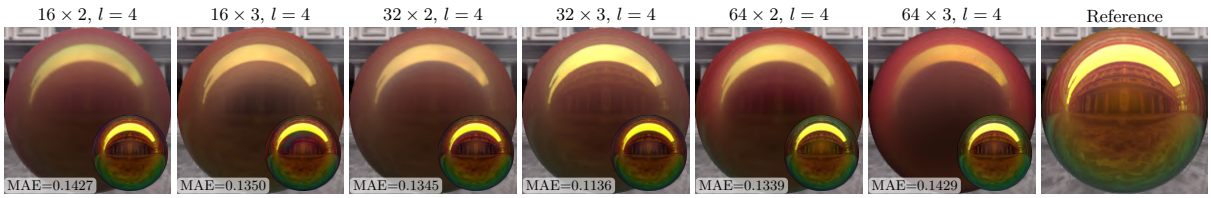

Figure 12: Our model struggles to reproduce complex phenomena such as iridescence (from `cc_amber_citrine` from RGL) accurately without using a large neural network. At the bottom of each image is displayed, (**Left**) the color-averaged MAE and (**Right**) a map of the absolute error.